

Observation Width Above Weight One: Block Compression and the Residual Gap

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August 2026

Abstract

The flagship paper’s transcript-span theorem proves that any transcript-stable exact realizer of $\{Q = 0\}$ observes refinement values at directions spanning the input, so weight- $\leq w$ observations number at least $\text{wt}(x)/w$; the weighted-source WITT-FIFO rule attains the bound at $w = 1$, and the flagship claims no matching construction for $w > 1$. This note proposes a resolution: for every fixed w , applying the flagship realization theorem to a *block-induced instance* – partition the support of x into blocks of weight at most w , take the block indicator vectors as new coordinates, the Gram matrix of B on the blocks as the induced alternating form, and the block values of Q as the induced diagonal – yields a rule under the same F1–F3/N1–N2 contract with $(w_0, c) = (w, 1)$ whose distinct observations are exactly the $\lceil \text{wt}(x)/w \rceil$ block vectors. Polarization makes the induced diagonal reconstruct $Q(x)$ exactly, so exactness is inherited, and the transcript-span bound is attained up to the integer ceiling at every width. The proof is a three-step reduction to the flagship theorem and is recorded in full; it has not yet passed the repository’s adversarial review gate, and the residual questions – all minor – are listed at the end.

1 The bound and its attainment at width one

Notation and contract are the flagship’s [1]: $V = \mathbb{F}_2^n$ with standard basis $e_1, \dots, e_n$, B alternating, Q a quadratic refinement of B , input x loaded by a refinement-blind map (F1), each transition metered to at most c oracle queries at Hamming weight at most w_0 (F2), one declared outcome convention (F3), exactness uniform over the refinement torsor (N1) within budget (N2).

Theorem 1 (transcript-span lower bound). **[PROVED]** [1, transcript-span theorem] *Let the refinement family at fixed B be closed under addition of every linear functional. If a transcript-stable rooted result is exact at x , then $x \in \text{span } S(q, x)$; in the original frame, if every observed vector has weight at most w , then $\text{wt}(x) \leq |S(q, x)| w$.*

Theorem 2 (attainment at $w = 1$). **[PROVED]** [1, optimal observation support] *The weighted-source rule satisfies the contract at $(w_0, c) = (1, 1)$ and its distinct observations at input x are exactly the $\text{wt}(x)$ singleton directions $\{e_j : x_j = 1\}$.*

The flagship then records: for $w > 1$ the theorem is only a lower bound; no matching construction is claimed. This note supplies the candidate construction.

2 The block-induced instance

Definition 3 (block partition). Fix $w \geq 1$ and an input $x \in V$. Partition $\text{supp}(x)$ into blocks $Z_1, \dots, Z_k$ of size at most w by greedy chunking in coordinate order, so $k = \lceil \text{wt}(x)/w \rceil$. Write $z_i = \sum_{j \in Z_i} e_j$ for the block indicator vectors. The partition is computed from x alone.

Definition 4 (induced instance). On $V' = \mathbb{F}_2^k$ with standard basis $e'_1, \dots, e'_k$, define

$$B'(e'_i, e'_j) = B(z_i, z_j) \quad (i \neq j), \quad Q'\left(\sum_{i \in s} e'_i\right) = Q\left(\sum_{i \in s} z_i\right) \quad (s \subseteq \{1, \dots, k\}),$$

and load the all-ones input $\mathbf{1} = e'_1 + \dots + e'_k$.

Lemma 5 (the induced instance is well formed). **[STANDARD MATH]** B' extends to an alternating bilinear form on V' , Q' is a quadratic refinement of B' with diagonal $Q'(e'_i) = Q(z_i)$, and

$$Q'(\mathbf{1}) = Q(x).$$

Proof. B' is the Gram matrix of B on the z_i : it is symmetric, and its diagonal $B(z_i, z_i)$ vanishes because B is alternating, so B' is alternating. The map $s \mapsto \sum_{i \in s} z_i$ is $\mathbb{F}_2$ -linear and injective (the z_i have pairwise disjoint nonempty supports), so Q' is the pullback of Q along a linear embedding; pullbacks of quadratic maps are quadratic, and the polar form of the pullback is the pullback of the polar form, which is B' by definition. Explicitly, iterating $Q(u + v) = Q(u) + Q(v) + B(u, v)$ gives

$$Q\left(\sum_{i \in s} z_i\right) = \sum_{i \in s} Q(z_i) + \sum_{\substack{i < j \\ i, j \in s}} B(z_i, z_j),$$

which is the polarization expansion of the quadratic map on V' with diagonal $Q(z_i)$ and polar B' . Since $z_1 + \dots + z_k = x$, evaluating at $s = \{1, \dots, k\}$ gives $Q'(\mathbf{1}) = Q(x)$. $\square$

Proposition 6 (block compression). **[PROVED] [AWAITING ADVERSARIAL REVIEW]** Fix $w \geq 1$. Running the weighted-source WITT-FIFO rule on the induced instance $(V', B', Q', \mathbf{1})$, with each induced singleton oracle query $Q'(e'_i)$ implemented as the original-frame query $Q(z_i)$, defines a rule satisfying F1–F3/N1–N2 at $(w_0, c) = (w, 1)$ whose root outcome is P iff $Q(x) = 0$, and whose distinct refinement observations are exactly the $k = \lceil \text{wt}(x)/w \rceil$ vectors $z_1, \dots, z_k$, each of original-frame weight at most w . These observations have pairwise disjoint supports, are linearly independent, and sum to x ; the transcript-span bound of Theorem 1 is therefore attained up to the integer ceiling at every width w .

Proof. Contract. F1: the block partition, the vectors z_i , the Gram data B' , and the loaded input $\mathbf{1}$ are computed from (B, x) and the fixed frame only; the flagship rule's own statics on the induced instance consult only $(B', \mathbf{1})$ and are refinement-blind by the flagship's locality proposition. F2: the flagship rule queries its oracle only at induced singletons e'_i , at most once per transition; each such query is implemented as one original-frame query at z_i , which has Hamming weight $|Z_i| \leq w$. Thus $(w_0, c) = (w, 1)$, constants independent of n . F3: the semantics is the flagship's single normal-play claim convention, unchanged. N1: for every refinement Q of every B and every x , Lemma 5 produces a refinement Q' of B' , and the flagship theorem is uniform over all refinements of all alternating forms in all finite dimensions, so it applies to $(V', B', Q', \mathbf{1})$; N2 holds with the stated budget.

Exactness. By the flagship realization theorem, the root of the induced arena is P iff $Q'(\mathbf{1}) = 0$, and $Q'(\mathbf{1}) = Q(x)$ by Lemma 5.

Observation support. By the flagship’s locality proposition, the distinct observations of the induced run at input $\mathbf{1}$ are exactly the induced singletons $\{e'_i : \mathbf{1}_i = 1\}$, i.e. all of them; under the implementation these are exactly $z_1, \dots, z_k$. Disjoint supports give independence, and $\sum_i z_i = x$ by construction. Finally $k = \lceil \text{wt}(x)/w \rceil$, against the lower bound $\text{wt}(x)/w$. $\square$

Remark 7 (total observed weight). The same construction is exact for the total-weight measure: the span requirement forces $\sum_{z \in S} \text{wt}(z) \geq \text{wt}(x)$ for any exact transcript-stable rule (the supports must cover $\text{supp}(x)$), and the block rule achieves $\sum_i \text{wt}(z_i) = \text{wt}(x)$ exactly. So neither the count-at-width nor the total-weight measure separates rules from certificates once Proposition 6 stands.

3 Status and residual questions

Proposition 6 is a new synthesis: its only inputs are the flagship realization and locality theorems (proved, with their Lean- checked ingredients) and the polarization expansion (standard). It has not yet been adversarially reviewed, and the flagship should not absorb it before that gate. Points a referee should press:

1. that the contract’s weight metering is genuinely frame-relative as used here – the flagship states its lower bound “in the original frame,” and the block rule’s queries are original-frame points of weight $\leq w$, but the F2 clause should be re-read against the blocked implementation;
2. that a per-input block partition is admissible under F1 – the loading map may consult x and B but not Q ; greedy chunking consults only $\text{supp}(x)$;
3. that N1 uniformity survives the composition – the induced family $\{(B', Q')\}$ ranges over all alternating forms with all refinements as (B, Q, x) ranges, and the flagship theorem quantifies over exactly that family.

If the proposition survives, the flagship’s second open question closes positively and the residual gap is only the integer ceiling $\lceil \text{wt}(x)/w \rceil$ versus $\text{wt}(x)/w$, which is not a real question. Two sharper readings would remain open:

Question 8 (uniform-in-width interface). **[OPEN]** Is there a single rule, one arena family with one access declaration, that attains the bound simultaneously at every width – e.g. whose observation set at declared width w is the block support – rather than one rule per fixed w ?

Question 9 (non-partition geometries). **[OPEN]** The block rule observes a disjoint-support family. Do overlapping-support observation geometries buy anything – for instance, exactness with fewer than $\lceil \text{wt}(x)/w \rceil$ queries on inputs with special polar structure – or does the span-plus-cover argument already force the partition count on every input? (For the uniform torsor guarantee the lower bound applies verbatim; the question is only about per-instance improvements below uniformity.)

This note is the authoritative statement of the width problem and will be folded into the flagship paper – as a corollary beside the $w = 1$ attainment if Proposition 6 survives review, as a retracted attempt otherwise.

References

- [1] a9lim, *Quadratic refinements in normal play: realization and the observation boundary*, companion flagship paper, `writeups/goldarf.tex` in the ogdoad repository, 2026.
- [2] C. Carlet, *Boolean Functions for Cryptography and Coding Theory*, Cambridge University Press, 2021.
- [3] Z. Zhang, Y. Shi, *On the parity complexity measures of Boolean functions*, arXiv:1004.0436 (2010).